PROJECT PROPOSAL · LARGE LANGUAGE MODELS AND ITS APPLICATIONS (83578-01)

DischargeLogic

LLM-Guided Feature Discovery for Interpretable Readmission Prediction

An LLM proposes and extracts clinical yes/no features from discharge notes; a Tsetlin Machine turns them into readable rules.

Harel Shrot · Alon Bezalel

Lecturer: Dr. Alexander Apartsin · Bar-Ilan University · November 2026

1 · MOTIVATING USE CASE

Hospital readmissions are frequent, costly and penalized

3.8M

30-day all-cause adult readmissions in the US in 2018

14 in 100

index admissions were followed by a readmission

$15,200

average cost of a single readmission

Since 2012, Medicare's Hospital Readmissions Reduction Program cuts payments by up to 3% for excess readmissions; 83% of hospitals were penalized in fiscal 2020.

Who acts on it: clinicians and discharge planners deciding who gets an early follow-up, a phone call or home care, and they need to know *why* a patient is flagged.

Sources: AHRQ HCUP Statistical Brief #278 (2018 data); Lu et al., BMC Health Services Research, 2021

1 · HOW IT IS HANDLED TODAY

Current tools trade transparency against what they can read

Clinical risk scores

LACE, HOSPITAL: a few hand-picked structured variables.

Transparent, but blind to what is only written in the notes.

Text models

ClinicalBERT reads the discharge summary: AUROC 0.714 on MIMIC-III.

Opaque: no reason a clinician can check.

Zero-shot LLMs

Flexible, no training needed.

Opaque, and can fail silently: at chance with a one-class bias on a credit-risk task.

Clinicians tend to favour simple models with interpretable predictors (McInerney et al., 2023), and high-stakes decisions call for interpretable models rather than explained black boxes (Rudin, 2019).

Sources: Huang et al., 2019 (ClinicalBERT); Butler et al., 2026; McInerney et al., 2023 (CHiLL); Rudin, 2019

2 · PROJECT TASK

Problem statement

> Given a patient's discharge summary, develop a system that predicts 30-day readmission and explains it with short rules over clinically meaningful yes/no features, while all patient data stays on local servers.

Why it is hard: the feature space is huge, and there are no obvious candidates

Anything can be a feature

Diagnoses, medications, social situation, follow-up plans...

Too many to extract

Each feature costs an LLM pass over ~330,000 notes

Risk in combinations

Weak findings may matter together, beyond one linear weight

The Tsetlin Machine: classification by voting rules

Binary input
heart failure = 1
lives alone = 1
follow-up = 0

→

Literals
Each feature and its negation: x and ¬x

→

Clauses
AND of chosen literals, e.g. HF ∧ alone ∧ ¬follow-up

→

Vote
Clauses for (+) minus clauses against (–) → class

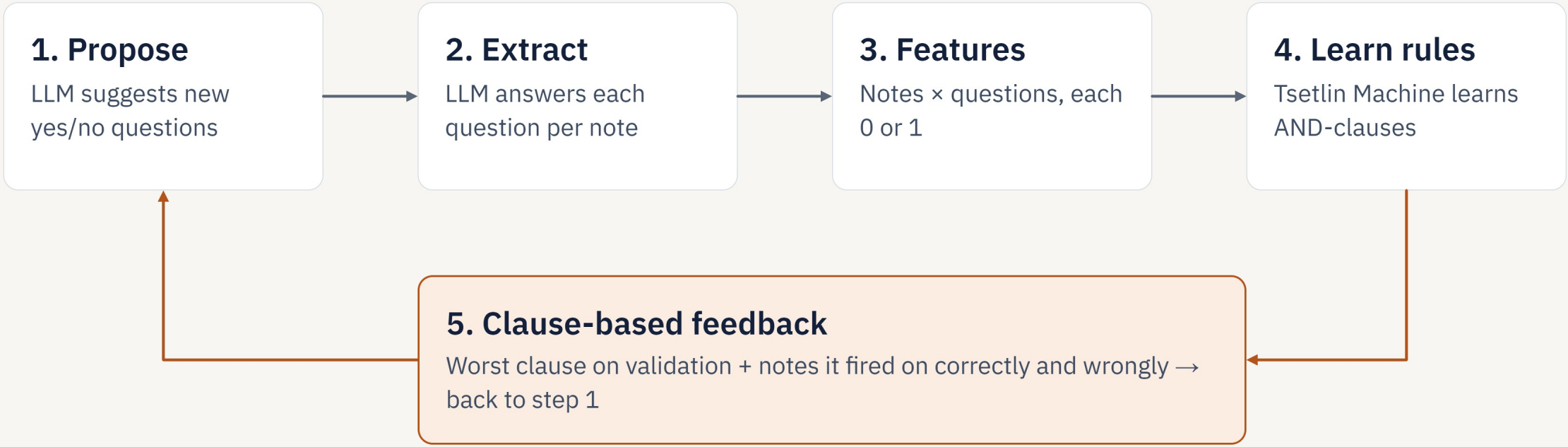
Learning: for each literal in each clause, a small automaton learns by reward and penalty whether to include it. No gradients, and retraining is fast.

Result: the trained model is a readable set of AND-rules.

Granmo, 2018 · illustrative features

3 · METHOD

A closed loop between an LLM and a Tsetlin Machine



Rules: training notes only · stop when validation AUROC stops improving or after a preset maximum of rounds · duplicate questions filtered · test set never touched. **After the loop:** distil the extractor into a small fine-tuned LLM.

3 · THE KEY STEP

Clause-based feedback: show the LLM the rule, not just the errors

MISFIRING CLAUSE

heart failure
∧ lives alone

votes for readmission

Fired correctly · readmitted

"...CHF exacerbation. Lives alone. To call primary care as needed..."

"...HF, EF 30%. Lives by herself. No appointments scheduled..."

Fired wrongly · not readmitted

"...CHF. Lives alone. Home nursing arranged; cardiology clinic in 5 days..."

"...heart failure, lives alone. Follow-up visit booked for next week..."

New question from the LLM: "Was a follow-up appointment scheduled after discharge?"

Illustrative synthetic example; no MIMIC text is shown. Existing loops give the LLM only misclassified samples and per-feature statistics.

3 · WHERE THE EXPLANATION COMES FROM

The model itself is the explanation

Global: the rules

heart failure ∧ lives alone ∧ ¬ follow-up booked → readmit

The whole model is a list of such clauses a clinician can read.

Local: one patient

Input: "...CHF. Lives alone. Call primary care as needed..."

↓ heart failure: yes · lives alone: yes · follow-up: no

↓ 3 clauses for, 1 against → high risk, with those clauses as the reason

Feature importance

How often a question appears in clauses, weighted by their votes.

Compare: BERT + SHAP approximates a black box after the fact. Logistic regression on the same questions gives one weight per question, with no combinations.

Illustrative synthetic example, not a result.

4 · NOVELTY AND CONTRIBUTION

What exists, and what we add

Already exists

LLM feature-discovery loops with a linear classifier (Butler et al., 2026; TBM, Ludan et al., 2023); static LLM-extracted features from clinical notes (CHiLL, 2023).

What we reuse

pyTsetlinMachine; a large open-weight LLM run locally; MIMIC-IV-Note and published splits; Hugging Face with LoRA.

Limitation that remains

Linear models cannot combine features, so the LLM writes ANDs into single features by hand; feedback is per feature; no Tsetlin Machine in such a loop.

What we add

A TM in the loop, so rules are learned conjunctions; clause-level contrastive feedback; quantitative tests of explanation quality; a distilled small extractor.

The pipeline’s components

Component	Choice	Why, and alternatives
Proposer and annotator	Large open-weight LLM run locally (about 300B parameters)	Strong open model; MIMIC text may not reach external APIs such as GPT
Rule learner	Tsetlin Machine (pyTsetlinMachine)	Learns AND-rules; retrains fast each round. Alternatives: logistic regression, decision tree
Trained model	Small LLM (1–3B), LoRA fine-tuned on the large LLM’s answers	Cheap local extraction

5 · DATA SPECIFICATION

MIMIC-IV discharge summaries

Source	MIMIC-IV-Note v2.2 + MIMIC-IV (hosp); Beth Israel Deaconess, Boston; de-identified; credentialed access via PhysioNet
Size	331,794 discharge summaries from 145,915 patients
Label	Next admission within 30 days of discharge (admissions table), using only the discharge note; in-hospital deaths excluded; about 17% positive
Split	As in Almeida et al. (2025): 60/20/20 by patient, stratified by label. The 20% test set is never used by the loop or for tuning

Risks: imbalance → AUROC and PR-AUC · leakage → split by patient · noisy LLM answers → hand-checked sample · privacy → local models only · readmissions to other hospitals not recorded → stated limitation

Sources: Johnson et al., MIMIC-IV-Note (PhysioNet, 2023); Almeida et al., 2025

6 · EVALUATION

Prediction and each component

Prediction

AUROC (primary) and PR-AUC, on the Almeida et al. split.

Reference on this cohort: 0.699 / 0.727 (Almeida et al.). ClinicalBERT 0.714 is MIMIC-III only.

Baselines: TM on words; logistic regression on the same features; LACE score.

Components

LLM extraction accuracy on a hand-checked sample.

Small vs large LLM: agreement, AUROC, cost.

Main ablation: AUROC per round, clause feedback vs error-only feedback.

We do not aim to beat state of the art; accuracy has to be comparable, and the focus is explainability.

Anticipated errors: the LLM misses negation (“no history of heart failure” answered yes) · a follow-up mentioned in another section of the note is missed · overly specific clauses that fit only a few training patients.

6 · MEASURING EXPLANATION QUALITY

Are the rules real and important?

Known ground truth

A label planted from structured MIMIC fields (e.g. diabetes code ∧ insulin order), independent of the LLM. Does the loop find matching questions in the notes?

Faithfulness

ERASER-style: remove, or keep only, the features the rules use. How much does the prediction change?

Agreement with other XAI

TM feature importance vs SHAP and tree importance, on the same features.

Clinical plausibility

Overlap with known risk factors (LACE: length of stay, acuity, comorbidity, ED visits).

Success: planted rule recovered · removing rule features cuts AUROC far more than random ones · short, readable rules · accurate extraction · AUROC comparable to Almeida et al.

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